

Jason Samuel Lucas

Assistant Professor, Department of Information Science
Director, Secure and Ethical AI Lab (SEAL)
College of Communication, Media, Design, and Information (CMDI)
University of Colorado Boulder, Boulder, CO 80309

(954) 864-4944
jason.lucas@colorado.edu
<https://www.jasonslucas.com>
<https://linkedin.com/in/jslu>

Appointments

University of Colorado Boulder, Boulder, CO

Assistant Professor, Department of Information Science, College of Media, Communication and Information,
Aug. 2026 – Present

Director, Secure and Ethical AI Lab (SEAL), Aug. 2026 – Present

Johns Hopkins University, Whiting School of Engineering, Baltimore, MD

Senior Researcher, Center for Language and Speech Processing, JSALT 2026 Workshop on Uncertainty-Aware
Health Coaching, Jun. 2026 – Aug. 2026

The Pennsylvania State University, University Park, PA

Doctoral Researcher, Amazon Nova AI Challenge (Team Lion-0xA), Feb. 2026 – Present

Graduate Research Assistant, PIKE Lab (PI: Dongwon Lee), Aug. 2021 – May 2026

NSF NRT LinDiv Fellow, Center for Language Science (PI: Janet G. van Hell), Aug. 2022 – Aug. 2024

Lawrence Livermore National Laboratory, Livermore, CA

DSSI Graduate Research Intern, Data Science Institute, May 2025 – Aug. 2025

SoundHound AI (Interactions LLC), New Providence, NJ

NLP Research Intern, May 2024 – Aug. 2024

Coalfire Systems, Greenwood Village, CO

Cybersecurity Consultant Intern, May 2023 – Aug. 2023

St. George's University, School of Medicine, St. George's, Grenada

Adjunct Assistant Professor and Instructor, IT, Medicine, and Public Health, Aug. 2010 – Jul. 2021

Head of B-Line Division, Clinical Skills Tech Lead, & SimCapture and ExamSoft Platform, 2012 – 2021

The Anglican High School, Government of Grenada, St. George's, Grenada

Certified Teacher II, Information Technology (98% CSEC Pass Rate), 2010 – 2012

Education

The Pennsylvania State University, University Park, PA

Ph.D. in Informatics (Data Science and AI/NLP), 2026

Dissertation: *Advancing Trustworthy AI Through Low-Resource Multilingual Natural Language Processing and Cybersecurity*

Advisor: Dr. Dongwon Lee

NSF NRT Graduate Research Traineeship Certificate, Linguistics and Language Sciences, 2024

St. George's University, St. George's, Grenada

M.P.H. in Epidemiology, 2020. *Chancellor's List for Academic Excellence.*

Advisor: Dr. Emanuel Keku.

Boston University, Boston, MA

M.Sc. in Computer Information Systems, Health Informatics concentration, 2014

Advisor: Dr. Vladimir Brusic.

St. George's University, St. George's, Grenada

B.Sc. in Information Technology, *Cum Laude*, 2010. *Dean's List and Chancellor's List*.

Research Interests

My research advances trustworthy and equitable AI for the world's languages and communities, with emphasis on low-resource and Caribbean linguistic contexts. The Secure and Ethical AI Lab (SEAL) at CU Boulder pursues four interconnected directions:

- Multilingual NLP and Low-Resource AI
- Trustworthy AI, AI Safety, and Information Integrity
- Fair, Ethical, and Equitable AI
- Human-Center AI

Distinctions, Fellowships & Awards

Social Impact Paper Award, Association of Computational Linguistics, Main Conference San Diego	2026
Research Excellence Ph.D. Award, Penn State College of IST	2026
NSF NRT Graduate Research Fellowship in Linguistic Diversity (LinDiv)	2022 – 2024
Funds for Excellence in Graduate Recruitment Fellowship, Penn State	2021
Pan-African Professional Alliance Leadership Award, Penn State	2024
Future Faculty Career Exploration Program Scholarship, Rochester Institute of Technology	2025
CRA-WP Grad Cohort for IDEALS Scholarship	2023, 2024, 2025
ICML and Black in AI Diversity & Inclusion Scholarship	2022
ACL Diversity & Inclusion Scholarship	2022
ACM Richard Tapia Conference Scholarship	2024
THRIVE BLK Men in Tech Conference Scholarship	2024
AAAI-22 Scholarship	2022
Graduate Faculty Fellowship, St. George's University	2018
Excellence Award for Objective Structured Clinical Examination, St. George's University	2012
Undergraduate Student Scholarship, St. George's University	2007

Refereed Journal Articles

Author name in **bold**.

1. Dominik Macko, Aashish Anantha Ramakrishnan, **Jason Lucas**, Robert Moro, Ivan Srba, Adaku Uchendu, and Dongwon Lee. 2026. Beyond Speculation: Measuring the Growing Presence of LLM-Generated Texts in Multilingual Disinformation. *IEEE Computer*.
2. **Jason Lucas**, Barani Maung Maung, Maryam Tabar, Keegan McBride, and Dongwon Lee. 2024. The Long-Tail Impact of Generative AI on Disinformation: Harmonizing Dichotomous Perspectives. *IEEE Intelligent Systems*.
3. Barani Maung Maung, Keegan McBride, **Jason Lucas**, Maryam Tabar, and Dongwon Lee. 2024. Generative AI Disproportionately Harms Long-Tail Users. *IEEE Computer* 57, 11.
4. Pureheart Ogheneogaga Irikefe, Reuben George, Peterson Ozili, Mohammad Falahat, Peter Nwachukwu Mba, Denis Nfor Yuni, and **Jason Lucas**. 2026. Predicting Sustainable Development Goals with Global AI Readiness: A

Hybrid Bayesian Belief Network and Artificial Neural Network Approach. *Journal of Open Innovation: Technology, Market, and Complexity* (Q1).

5. Pureheart Ogheneogaga Irikefe, Mohammad Falahat, Kings Ferrels Kondowe, **Jason Lucas**, and Kwame Ntim Sekyere. 2026. Cybernetic Principles and Artificial Neural Networks in Social Science Scale Development: A Systematic Review and Science Mapping. *Cogent Social Sciences* (Q1).

Book Chapters

Author name in **bold**.

1. **Jason Lucas**, Pureheart Ogheneogaga Irikefe, Mohammad Falahat, and Jonathan Ojarikre Oniore. 2026. Sustainable Development Attitudes and Pro-Environmental Behaviour in ASEAN: A Systematic Review and Meta-Analysis. In *Business of the Future: Innovating Sustainability*. Springer.

Refereed Conference Articles

1. Meruyert Aristombayeva, **Jason S. Lucas**, Chaewan Chun, and Dongwon Lee. 2026. Lost in Speech: Trilingual Spoken Hallucination Detection Across Audio and Transcripts. In *Proceedings of the 2nd Workshop on Speech and Audio Language Models (SALMA 2026)*, co-located with the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP '26), Oct. 24–29, 2026, Budapest, Hungary.
2. Jiayi Yang, Chaewan Chun, **Jason Lucas**, Yuchen Yang, and Dongwon Lee. 2026. AOR-Bench: Do Large Audio Language Models Over-Refuse Pseudo-Harmful Queries? In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP '26)*, Oct. 24–29, 2026, Budapest, Hungary.
3. **Jason Lucas**, Matt Murtagh-White, Adaku Uchendu, Ali Al-Lawati, Michiharu Yamashita, Dominik Macko, Ivan Srba, Robert Moro, and Dongwon Lee. 2026. BLUFF: Benchmarking Falsehoods in Long-Tail Multilingual Low-Resource Languages. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26)*, Datasets and Benchmarks Track, Aug. 9–13, 2026, Jeju, South Korea.
4. Ali Al-Lawati, **Jason Lucas**, Dongwon Lee, and Suhang Wang. 2026. Position: LLM Benchmark Datasets Should Be Contamination-Resistant. In *Proceedings of the 43rd International Conference on Machine Learning (ICML '26)*, Position Paper Track, Jul. 13–19, 2026, Seoul, South Korea.
5. **Jason Lucas**, Pureheart Irikefe, Adaku Uchendu, Umniya Najaer, Cornelius Adejoro, Patrice Sterling, and Dongwon Lee. 2026. Position: Breaking the Dual Curse of Multilingual AI Requires Socio-Technical Guardrails, Not Post-Hoc Alignment. In *Proceedings of the 43rd International Conference on Machine Learning (ICML '26)*, Position Paper Track, Jul. 13–19, 2026, Seoul, South Korea.
6. **Jason Lucas**, Matt Murtagh, Ali Al-Lawati, Adaku Uchendu, and Dongwon Lee. 2026. DIA-HARM: Dialectal Disparities in Harmful Content Detection across 50 English Dialects. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL '26)*, San Diego, CA, USA. (**Social Impact Paper Award**)
7. **Jason Lucas**, John Chen, Ali Al-Lawati, Mahjabin Nahar, and Mahnoosh Mehrabani. 2025. Chain-of-Interactions: Iterative ICL Framework for Abstractive Task-Oriented Dialogue Summarization of Conversational AI Interactions. In *Findings of the Association for Computational Linguistics: EMNLP 2025*.
8. Ali Al-Lawati, **Jason Lucas**, Zhiwei Zhang, Prasenjit Mitra, and Suhang Wang. 2025. Graph-Based Molecular In-Context Learning Grounded on Morgan Fingerprints. In *Findings of the Association for Computational Linguistics: EMNLP 2025*.
9. Ekaterina Artemova, **Jason Lucas**, Saranya Venkatraman, Jooyoung Lee, Sergei Tilga, Adaku Uchendu, and Vladislav Mikhailov. 2025. Beemo: Benchmark of Expert-Edited Machine-Generated Outputs. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL '25)*, pages 6992–7018.
10. Ali Al-Lawati, **Jason Lucas**, and Prasenjit Mitra. 2025. Semantic Captioning: Benchmark Dataset and Graph-Aware Few-Shot In-Context Learning for SQL2Text. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING '25)*, pages 8026–8042, Abu Dhabi, UAE.

11. Dominik Macko, Robert Moro, Adaku Uchendu, Ivan Srba, **Jason Lucas**, Michiharu Yamashita, Nafis Irtiza Tripto, Dongwon Lee, Jakub Simko, and Maria Bielikova. 2024. Authorship Obfuscation in Multilingual Machine-Generated Text Detection. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 13706–13724.
12. Dominik Macko, Robert Moro, Adaku Uchendu, **Jason Lucas**, Michiharu Yamashita, Matúš Pikuliak, Ivan Srba, Thai Le, Dongwon Lee, Jakub Simko, and Maria Bielikova. 2023. MULTITuDE: Large-Scale Multilingual Machine-Generated Text Detection Benchmark. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP '23)*, pages 9960–9987, Singapore.
13. **Jason Lucas**, Adaku Uchendu, Michiharu Yamashita, Jooyoung Lee, Shaurya Rohatgi, and Dongwon Lee. 2023. Fighting Fire with Fire: The Dual Role of LLMs in Crafting and Detecting Elusive Disinformation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP '23)*, pages 14279–14305, Singapore.
14. **Jason Lucas**, Limeng Cui, Thai Le, and Dongwon Lee. 2022. Detecting False Claims in Low-Resource Regions: A Case Study of Caribbean Islands. In *Proceedings of the Workshop on Combating Online Hostile Posts in Regional Languages during Emergency Situations (CONSTRAINT, ACL '22)*, pages 95–102, Dublin, Ireland.

Manuscripts Under Review and In Preparation

1. **Jason Lucas**, Umniya Najaer, and Dongwon Lee. 2026. When AI Rewrites Itself: Auditing Alone Cannot Govern Self-Modifying AI.
2. **Jason Lucas**, Ali Al-Lawati, Matt Murtagh-White, Md Toufikuzzaman, Rajveer Dharkar, Adaku Uchendu, Yuchen Yang, Xinyu Xing, and Dongwon Lee. 2026. DIA-Guard: Dialectal Evasion Attacks and Defenses Against Safety-Aligned AI Models.
3. Meruyert Aristombayeva, **Jason Lucas**, Chaewan Chun, and Dongwon Lee. 2026. Lost in Speech: Trilingual Spoken Hallucination Detection Across Audio and Transcripts.
4. Ali Al-Lawati, Nafis Tripto, Abolfazl Ansari, **Jason Lucas**, Suhang Wang, and Dongwon Lee. 2026. Ask Before You Block: Intent-Aware Moderation in MoltBook.

Dissertation, Theses & Capstone

1. **Jason S. Lucas**. 2026. *Advancing Trustworthy AI Through Low-Resource Multilingual Natural Language Processing and Cybersecurity*. Ph.D. Dissertation. College of Information Sciences and Technology, The Pennsylvania State University, University Park, PA, USA. May 2026. Advisor: Dongwon Lee.
2. **Jason S. Lucas**. 2020. *Effects of Self-Care Apps in Prediabetics in Grenada*. M.P.H. Capstone. School of Graduate Studies, St. George's University, St. George's, Grenada. Aug. 2020. Advisor: Emanuel Keku.
3. **Jason S. Lucas**. 2014. *Advancing Health Information Systems (EHR) in Low- and Middle-Income Countries*. M.Sc. Capstone. Metropolitan College, Boston University, Boston, MA, USA. Sep. 2014. Advisor: Vladimir Brusic.
4. **Jason S. Lucas**. 2010. *Azure Medical Clinic Web Application*. B.Sc. Information Systems Project. Department of Computers and Technology, St. George's University, St. George's, Grenada. May 2010. *Cum Laude*. Advisor: Michael Roberts.

Conference Oral Presentations

1. **2026** The 64th Annual Meeting of the Association for Computational Linguistics (ACL), San Diego, CA, USA. *DIA-HARM: [full paper title]*.
2. **2023** The 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), Sentosa, Singapore. *Fighting Fire with Fire: The Dual Role of LLMs in Crafting and Detecting Elusive Disinformation*.
3. **2022** The 60th Annual Meeting of the Association for Computational Linguistics, CONSTRAINT Workshop, Dublin, Ireland. *Detecting False Claims in Low-Resource Regions: A Case Study of Caribbean Islands*.

Invited Talks and Presentations

Schloss Dagstuhl – Leibniz-Zentrum für Informatik , Wadern, Germany	Jun. 2026
<i>Dagstuhl Seminar 26252: From Speech Translation to Multilingual Communication.</i>	
St. George's University , LEAD Faculty Workshop, Grenada	2026
<i>Responsible and Ethical Use of Generative AI in Teaching and Research.</i>	
Munich NLP , Munich, Germany	2025
<i>Beemo: Benchmark of Expert-Edited Machine-Generated Outputs.</i>	
Penn State University , Startup Week Panel, University Park, PA	Mar. 2025
<i>Deepfake for Social Good and Applications for Patients with Aphasia.</i>	
Penn State College of IST , Research Panel, University Park, PA	Jan. 2025
<i>Cutting-Edge Research and Future Directions for Combating Harmful Text.</i>	
Cohere for AI , Virtual	May 2024
<i>Advancing Low-Resource Multilingual Approaches in the Era of Artificial Intelligence.</i>	
St. George's University , Department of Computers and Technology, Grenada	Jan. 2024
<i>Forum on Opportunities in Emerging Technology.</i>	
St. George's University , Department of Computers and Technology, Grenada	Nov. 2023
<i>Artificial Intelligence and Latest AI Research.</i>	
NSF and IES AI Research Institutes Meeting , University Park, PA	Sep. 2023
<i>AI Transforming Education for Children with Speech and Language Pathologies.</i>	
Pan-APA 6th Annual Conference , University Park, PA	Apr. 2023
<i>Detecting False Claims in Low-Resource Regions: A Caribbean Case Study.</i>	
CRA-WP Grad Cohort for IDEALS , Honolulu, HI	Mar. 2023
<i>Lightning Talk: Detecting False Claims in Low-Resource Regions.</i>	

Teaching Experience

University of Colorado Boulder

INFO 4615: Fair Machine Learning	Fall 2026
An introduction to fairness in machine learning: what fairness means (and doesn't), how to measure disparate impact, and how to intervene during data collection, model training, and deployment.	

The Pennsylvania State University

IST 597: Applied Generative AI	Spring 2025
Graduate Teaching Assistant. Led labs on LLM applications, prompt engineering, and in-context learning; designed assignments and assessments.	
IST 402: Posthuman Performance — Humans, Robotics, and AI in Action	Spring 2025
Graduate Teaching Assistant. Supervised undergraduate lab projects on Boston Dynamics Spot simulation and collaborative robotic arms.	
IST 210: Organization of Data	Spring 2022
Graduate Teaching Assistant. Database fundamentals and Microsoft SQL practical instruction.	
IST 140: Introduction to Application Development	Fall 2021
Graduate Teaching Assistant. Java programming using IntelliJ.	

St. George's University

BPM II: Basic Principles of Medicine — Biomedical Informatics	Spring 2021
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Lecturer. Developed course material, assessments, and exam questions.

PUBH 302: Public Health Informatics

2020 – 2021

Lecturer. Developed course material .

COMP 420: Database Systems

2010 – 2011

Adjunct Assistant Professor and Course Director.

The Anglican High School

Information Technology (Form 3-5)

[2010]

Information Technology Teacher and Form Head, Class 4S1. Taught IT to upper/senior high school students and managed the school's IT laboratory; prepared students for the CSEC (CXC) examination.

Mentorship and Advising

Doctoral and Postdoctoral

Recruiting Ph.D. students and postdoctoral researchers for the Secure and Ethical AI Lab (SEAL), Department of Information Science, College of Media, Communication and Information, CU Boulder.

Undergraduate Research Mentees

Gracy Juanita Franco Prasanna, Penn State — AI for Social Good using Deepfake	2025
Colby James Gleason, Penn State — Image Emotion and Sentiment Classification	2025
Pranav Jain, Penn State — Boston Dynamics Spot Robot Simulation	2025
Michael Wayes Charles, Penn State — Data Sensorium	2025
Kendall Reed II, Penn State Millennium Scholar — Multilingual NLP, PIKE Lab	2023 – Present
Lia Carin Djaouga, Penn State Millennium Scholar — Multilingual NLP, PIKE Lab	2023 – Present

Independent Research Mentees

Rajveer Dharkar, West Windsor-Plainsboro High School North — AI/NLP	2025 – Present
Elvis Tata Tanghang, Carnegie Mellon University — AI/NLP	2025 – Present
Farzana Islam, Independent University, Bangladesh — AI/NLP	2025 – Present

Professional Service

Area Chair

ARR ACL 2026: 64th Annual Meeting of the Association for Computational Linguistics

ARR EMNLP 2026: 31st Conference on Empirical Methods in Natural Language Processing

Conference Program Committee Member & Reviewer

ACL 2025, 2026

EMNLP 2024, 2025

NeurIPS 2026

EACL 2026

KDD 2026 (Datasets and Benchmarks Track)

Workshop Program Committee Member

LoResLM 2026: Workshop on Low-Resource Languages and Multilingual NLP, co-located with EACL
Workshop on Detecting AI-Generated Content, co-located with COLING 2025

Conference and Organizing Roles

Organizing Committee, 12th Mid-Atlantic Student Colloquium on Speech, Language, and Learning 2025
Conference Volunteer, EMNLP 2023 2023

Departmental and Institutional Service

Graduate Committee, Department of Information Science, CU Boulder 2026 – Present
Manage Master’s program admissions and department website.
CMDI Faculty Council Representative, Department of Information Science, CU Boulder 2026 – Present
Serve as INFO Representative
Graduate Student Bill of Rights Committee, College of Info. Sc. & Tech, Penn State 2024
President, Graduate Student Association in the College of Info. Sc. & Tech, Penn State 2022 – 2023
IT Officer, Penn State Pan-African Professional Alliance, Penn State 2023 – 2025
Outreach Coordinator, NSF NRT LinDiv Program, Penn State 2022 – 2023

Mentorship and Outreach

Faculty Mentor, Penn State Cybersecurity Consulting Club Industry Trek to RSM and PwC, NYC Apr. 2026
Faculty Mentor, BLK Men in Tech THRIVE ’24 Conference, Penn State Delegation, Miami, FL 2024
Instructor, ENVISION STEM Career Day for Young Women, Penn State 2022 – 2023
Director and Co-founder, Fit For Life Youth Development Program, Grenada 2010 – 2017
IT Officer, Grenada Red Cross 2020 – 2021

Datasets and Benchmarks

DIA-HARM: Dialectal Disparities in Harmful Content Detection across 50 English Dialects (ACL ’26) 2026
<https://github.com/js15710/dia-harm>
DIA-Guard: AI Safety Evaluation and Defense Against English Dialectal Variational Attacks 2026
<https://github.com/js15710/dia-guard>
SHIELD: Safety Evaluation Benchmark for Dialectal Evasion Attacks Against Safety-Aligned AI Models 2026
<https://huggingface.co/datasets/js15710/Shield>
BLUFF: Benchmarking Falsehoods in Long-Tail Multilingual Low-Resource Languages 2026
<https://huggingface.co/datasets/js15710/BLUFF>
Caribbean False Claims Dataset (Lucas et al., CONSTRAINT @ ACL ’22) 2022

Press & Media Coverage

CMDI Now, University of Colorado Boulder, “Heads of the Class” Aug. 2026
Slator, “New Benchmark Tests AI Detection Across Languages and Translation” Mar. 2026
Penn State University, “IST and Engineering Students Competing in Global Amazon Nova AI Challenge” 2026
St. George’s University, “From SGU to the Frontiers of AI: Jason Lucas’ Journey of Purpose” 2025
Penn State IST iConnect, “Leadership Examples in Tech Rarely Look Like Me” 2025
Penn State News, “Penn State Connections Lead Doctoral Student to Interdisciplinary College of IST” 2025

WXXI News, “Artificial Intelligence in Journalism: How Is It Being Used Now? What’s Coming Next?” 2024

Technical Skills

Programming Languages	Python, Java, SQL, LaTeX, HTML5, CSS, PHP, R, C++
Machine Learning	PyTorch, TensorFlow, Hugging Face Transformers, Adapter-Transformers, AdapterHub, vLLM, Pandas, NumPy, scikit-learn
Data Collection	Twitter API, Selenium, Tesseract
Platforms & Tools	VS Code, PyCharm, IntelliJ IDEA, Google Cloud, Microsoft Azure, Streamlit, MS Office, Epi Info, GeoDa
Domain Expertise	Health Informatics, Epidemiology, AI Engineering, Data Science Engineering, Health Information Systems (EHR)

Languages

English (Native), Grenadian English Creole (Native)

Professional Memberships

Association for Computational Linguistics (ACL)
 Association for Computing Machinery (ACM)
 Association for the Advancement of Artificial Intelligence (AAAI)

Research Links

GitHub	https://github.com/jsl5710
Google Scholar	https://scholar.google.com/citations?user=XU1WN6YAAAAJ
LinkedIn	https://www.linkedin.com/in/jslu
My Homepage	https://www.jasonslucas.com
Semantic Scholar	https://www.semanticscholar.org/author/Jason-Samuel-Lucas/2260072705

References

Dr. Andrea Tapia, Dean and Professor, College of IST, The Pennsylvania State University
 College Dean, Professor and Ph.D. Committee Member axh50@psu.edu

Dr. Dongwon Lee, Professor, Frymoyer Chair and HOD, College of IST, The Pennsylvania State University
 Ph.D. Advisor, Frymoyer Chair, Professor and Teaching Reference dongwon@psu.edu

Dr. Adaku Uchendu, Member of Technical Staff, MIT Lincoln Laboratory
 Ph.D. Mentor and Research Scientist adaku.uchendu@ll.mit.edu

Dr. Janet G. van Hell, Distinguished Professor of Psychology and Linguistics, Penn State
 LinDiv Advisor; Director, Center for Language Science jgv3@psu.edu

Dr. Betsy Campbell, Associate Teaching Professor, College of IST, Penn State
 Teaching and Mentorship Reference bjc28@psu.edu